Balram Maurya

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Professional Summary

Final-year Electronics and Communication Engineering student graduating in June 2027, seeking entry-level embedded systems, firmware and hardware roles. Hands-on experience with ESP32, Embedded C/C++, sensor interfacing, motor control and power integration through robotics and drone projects. Completed NEO, a four-wheel companion robot, as a final-year project. Brings practical prototyping, firmware debugging, bench testing, datasheet interpretation and technical documentation, with experience coordinating work across multidisciplinary teams.

Education

Thapar Institute of Engineering & Technology, Patiala Graduation: June 2027

Bachelor of Engineering, Electronics & Communication Engineering (final year)

Technical Skills

Programming & Firmware: Embedded C, C++, Python, ESP32 (32-bit MCU), Arduino, peripheral integration, sensor data acquisition, interrupt and timing handling
Interfaces & Peripherals: I2C, I2S, UART, GPIO, PWM, Wi-Fi, MPU6050/IMU, SSD1306 OLED, INMP441 microphone, MAX98357A audio amplifier
Hardware & Power: Prototype assembly, DC and brushless motors, motor drivers, electronic speed controllers (ESCs), Li-Po batteries, DC-DC buck regulation, wiring, power distribution
Testing & Tools: Git, VS Code, Arduino IDE, bench testing, continuity checks, calibration, firmware debugging, hardware troubleshooting, root-cause analysis, datasheet interpretation
Engineering Practices: Requirements analysis, component evaluation, hardware–software integration, system testing, test documentation, assembly notes, cross-functional collaboration

Embedded & Hardware Projects

NEO — Embedded Mobile Companion Robot Completed | Final-Year Project

- Developed ESP32 C++ firmware integrating an SSD1306 OLED over I2C, an INMP441 microphone and a MAX98357A amplifier over I2S for animated expressions and audio capture/playback.
- Designed and assembled a four-wheel platform with DC motors, motor-control hardware, Li-Po supply and DC-DC regulation for logic, audio and drive subsystems.
- Implemented Wi-Fi communication between the host application and ESP32 for wireless commands and responses; debugged connectivity and timing faults across device interfaces.
- Tested the integrated robot on the bench and resolved wiring, peripheral and power-stability issues through iterative hardware and firmware troubleshooting.

ESP32 Quadcopter Prototype — Sensor & Motor Integration First-Year Hardware Project

- Interfaced an MPU6050 IMU over I2C for accelerometer and gyroscope acquisition; configured four PWM motor-control channels for ESC response testing.
- Assembled the frame and integrated brushless motors, electronic speed controllers, Li-Po power distribution and a DC-DC regulator for the electronics supply.
- Validated sensor acquisition and motor-response logic through continuity checks, calibration and staged motor tests; troubleshot wiring, communication and power issues.

Autonomous Drone System — IRoC-U 2025 Team AGNIV; Round 2 selection

- Contributed to design and planning with a multidisciplinary team selected for Round 2 of the ISRO Robotics Challenge–University; supported sensor integration, data acquisition and system testing.
- Worked with Pixhawk, Raspberry Pi, LiDAR and IMU subsystems; evaluated component options against power, payload and reliability constraints and prepared requirement and test documentation.

Leadership & Project Coordination

Finance Secretary — Institution's Innovation Council (IIC), Thapar Institute 2025 – 2026

- Managed an INR 2L+ budget across 8+ initiatives, tracking allocations, expenses and reconciliations in Excel and preparing financial summaries for faculty review.
- Coordinated faculty, student teams and vendors to align requirements, follow up on procurement dependencies and resolve execution issues within agreed timelines.

Publishing Experience & Achievements

- Published 5+ books as Sterling Moriz, including 5 Proven Secrets to Crush Procrastination and Master AI Productivity (co-authored with Alex Ross); managed research, writing, editing and digital publishing.
- Innovation Ambassador (Level 2 — Advanced), Institution's Innovation Council, Ministry of Education; Participant, Axxelrate 2026.